

In-Context Learning with Recurrent Memory: State, Capacity, and Interference

DataForge 2026 × Pathway Track | Authoritative
Concept Briefing

Team Public Repo: github.com/LuciferK47/recurrent-memory-explainer

Core Papers: [arXiv:2509.26507](https://arxiv.org/abs/2509.26507) · [arXiv:2608.09888](https://arxiv.org/abs/2608.09888)

1. EXECUTIVE SUMMARY & LOCKED CLAIM

Autoregressive Transformers enable in-context learning via an explicit key–value (KV) cache scaling linearly with sequence length ($O(n)$ memory footprint per sequence), creating massive memory bandwidth and VRAM constraints. Fixed-size recurrent memory replaces unbounded buffers with a state matrix $M \in \mathbb{R}^{d \times d}$, achieving strictly $O(1)$ memory complexity during inference.

Locked Claim: “A fixed-size recurrent memory can absorb key–value associations from demonstrations and recall them accurately — until the number of associations exceeds the memory’s rank, at which point recall degrades through interference in a predictable, measurable way.”

2. ALGEBRAIC SUBSTRATE: HEBBIAN WRITE & READOUT

Associative memory updates via two primitive tensor operations:

- Outer-Product Write:** $M \leftarrow M + v \cdot k^T$ (Hebbian rank-1 write)
- Linear Readout:** $\hat{v} = M \cdot q = \sum_{i=1..N} v_i (k_i^T q)$

For query $q = k_j$, readout decomposes into signal and cross-talk:
 $\hat{v} = v_j (k_j^T k_j) + \sum_{i \neq j} v_i (k_i^T k_j) = v_j + \sum_{i \neq j} v_i (k_i^T k_j)$. If keys are mutually orthogonal ($k_i^T k_j = 0$), retrieval is exact.

3. THE CAPACITY CLIFF & INTERFERENCE

In $\mathbb{R}^d$, at most d orthogonal vectors exist. Once stored pairs $N > d$, cross-talk corrupts retrieval:

- Sub-capacity ($N \leq d$):** Cosine similarity ≈ 1.0 ; near-perfect recall.
- Critical Threshold ($N \approx d$):** Interference noise equals primary signal.
- Interference Regime ($N \gg d$):** Accuracy plunges toward $1/\sqrt{d}$ noise floor.

Interference is not a numerical error; it is the fundamental information-theoretic tax of bounded state compression.

4. ARCHITECTURAL COMPARISON

Architecture	Memory Footprint	State Update Rule	In-Context Learning Mechanism	Primary Bottleneck
Transformer (KV-Cache)	$O(n)$ sequence length	Append key & value token vectors	In-context lookup over token buffer	VRAM & memory bandwidth exhaustion
Infini-attention (arXiv:2404.07143)	$O(1)$ compressive block	$M \leftarrow M + \sigma(v) \cdot \sigma(k)^T$	Linear memory readout + local attention	State saturation over unbounded contexts
Titans (arXiv:2501.00663)	$O(1)$ neural memory module	Neural gradient test-time step	Learned test-time associative memory	Dual-system computational overhead
BDH / BDH-CQ (Pathway)	$O(1)$ synaptic state matrix	$M \leftarrow M + v \cdot k^T$ (Hebbian)	Zero-gradient demonstration absorption	Orthogonality capacity cliff ($N > d$)

5. ROLE OF BDH & BDH-CQ

- BDH (arXiv:2509.26507):** Biologically-inspired, scale-free graph of neuron particles where working memory emerges directly from Hebbian synaptic plasticity. Low-rank GPU formulation runs with ~5% activation sparsity and monosemantic wiring.
- BDH-CQ (arXiv:2608.09888):** Absorbs demonstration pairs (e.g. ARC-AGI grids) into recurrent state without backpropagation or chain-of-thought tokens, querying state to output solutions.

6. EVIDENCE DISCIPLINE & HONEST LIMITATIONS

- Self-Reported Benchmark 29.5% pass@2** on ARC-AGI-1 ($\$0.0007/\text{task}$) is audited by partner-institution co-authors, not an arm’s-length external lab. Scaling laws across multi-scale models remain unproven.
- Disclosed Failure:** BDH-CQ failed completely (0/72) on color-swap composition tasks, exposing vulnerability of pure outer-product states to relational logic.
- Didactic Scope:** Explainer runs live mathematical Hebbian updates in JS/TS to illuminate principles, without misrepresenting itself as the proprietary engine.